

First-Gate Support Leakage in MORNING-ATLAS-128: Exact s_1 Recovery and Fresh-Message Forgeries from Three Million Signatures per Key

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Provenance and review status. The attack was independently found by an AI system, OpenAI Codex using Daybreak Blue at maximum reasoning effort, without a human first proposing this specific attack. The derivation, implementation, experiments, and initial manuscript were also produced by the AI. At the date above, no human cryptanalyst has independently verified the argument, experiments, or novelty assessment. This is a preliminary disclosure for human review and reproduction.

Abstract

MORNING-ATLAS is a module-LWR signature submitted to the Next-generation Commercial Cryptographic Algorithms Program. ATLAS-128 samples every mask coefficient uniformly from $[-(\gamma_1 - 1), \gamma_1 - 1]$ and publishes $z = y + cs_1$ only when $\|z\|_\infty < \gamma_1 - \beta_1$. The specification argues that translating the accepted interval by a coefficient of cs_1 does not change its size. This overlooks truncation by the finite support of y . For ATLAS-128, $\beta_1 = 51$ while a challenge product can have magnitude $\kappa\eta = 496$. If $a = (cs_1)_j$, the number of masks compatible with an accepted response coefficient is exactly

$$1,048,473 - (|a| - 51)_+,$$

and boundary-near response coefficients supply exact public half-spaces containing s_1 .

Feasibility alone is insufficient because the zero vector satisfies all of these half-spaces. In an ideal first-gate model, however, the conditional likelihood of every feasible transcript is inversely proportional to the candidate's first-gate acceptance probability. A public bound-moment regression and radial linear programs put every target coefficient inside a public floor/ceiling box. Response-only integer programs give a feasible baseline. A deterministic polynomial-support ladder then combines 64 public LWR equations with each binary box in a Kannan embedding; LLL followed by BKZ-10 exposes s_1 exactly. A public-hint step recovers the omitted low part t_0 , and the recovered (s_1, t_0) forms an equivalent signing key.

On a calibration key, an execution on 3,000,000 ordinary chosen-message signatures retained 1,934,304 support rows, recovered all 768 coefficients, and produced a verifier-accepted fresh-message forgery. Because the completion method was developed retrospectively on that key, we then froze the code, parameters, public block order, and stopping rules before selecting a second key. With both ground-truth secret components withheld from the recovery and evaluation pipeline until the public result and forgery were sealed, the same procedure recovered exact (s_1, t_0) from a new three-million-signature transcript and forged a fresh message. On this prospective run, the response-only integer baseline was already exact; the prescribed lattice stage rediscovered it at block 0 after BKZ-10. Four-core collection took 5,207.6 and 5,064.7 seconds on the two keys, and peak public recovery memory was 1.45 GiB.

The corrected samplers target the written design, separately from the published two-signature repeated-mask implementation attack. Both tested keys succeeded, but two trials do not estimate a key-averaged success probability.

1 Introduction

Fiat–Shamir-with-aborts signatures publish a response obtained by adding a secret-dependent challenge product to a random mask. Rejection sampling is supposed to remove the dependence on the signing secret. A proof of independence must account for both the desired response interval and the finite support from which the mask was originally sampled.

MORNING-ATLAS follows this paradigm over a power-of-two cyclotomic ring [1]. Its specification samples y uniformly from a coefficient box and forms $z = y + cs_1$. The first signing gate retains z only inside a slightly smaller box. Section 2.5 of the specification states that shifting that smaller interval by a coefficient of cs_1 leaves exactly $2(\gamma_1 - \beta_1) - 1$ admissible mask values. That count is valid only when the shifted interval stays inside the sampler support. The ATLAS-128 parameters do not ensure this condition: $\beta_1 = 51$ and $\|cs_1\|_\infty$ can be as large as 496.

This mismatch has two consequences. First, a response near a mask boundary gives an exact linear inequality in s_1 . Second, challenges that produce larger products have lower first-gate acceptance probability. Once we condition on observing a valid signature, feasible candidate secrets with lower acceptance probability receive higher likelihood. A direct likelihood optimization is nonlinear. A Gaussian approximation suggests that the acceptance loss grows with the squared coefficient norm, leading to a simple sequence of public linear programs over the exact support polytope.

The public key supplies a decisive validation relation. If the full rounded product t is known, a candidate s_1 must satisfy

$$As_1 - 32t = e \pmod{2^{23}}, \quad e \in [-16, 15]^{9 \times 128}.$$

ATLAS omits the low part t_0 from its public key, but ordinary signature hints expose bounded observations of ct_0 . A public minimax computation on 20,000 signatures recovers all 1,152 coefficients of t_0 . The same messages are a subset of the three-million-signature response transcript, so the total number of distinct chosen-message queries remains three million.

Our contributions are:

- We give the exact accepted-support formula and identify the omitted truncation in the ATLAS rejection-sampling analysis.
- We derive exact public half-spaces and the ideal first-gate candidate likelihood. We clearly separate these statements from the norm-based heuristic used for efficient recovery.
- We specify a recovery procedure without access to the target secret, consisting of a public regression, radial cutting-plane linear programs, response-feasible integer completion, and a deterministic LWR lattice ladder with complete public validation.
- On two separately generated ATLAS-128 targets, the procedure recovers s_1 exactly, combines it with publicly recovered t_0 , constructs equivalent signing keys, and produces accepted signatures on fresh messages. The second execution was frozen before its key was generated and completed before the target secret was revealed to the recovery and evaluation process.
- We test the written design with corrected uniform samplers and distinguish the result from the previously published implementation attack caused by repeated adjacent mask coefficients [2].

The attack uses no timing information, fault, malformed signature, submitted sampler repetition, or access to signing randomness. It consumes fields decoded from signatures returned by an ordinary chosen-message signing oracle.

2 ATLAS-128 and the faithful target

Let

$$\mathcal{R}_q = \mathbb{Z}_q[X]/(X^{128} + 1), \quad q = 2^{23}, \quad p = 2^{18}.$$

ATLAS-128 uses a public matrix $A \in \mathcal{R}_q^{9 \times 6}$ and a secret $s_1 \in \mathcal{R}_q^6$ whose 768 coefficients are independently uniform in $[-\eta, \eta]$ with $\eta = 16$. Key generation rounds As_1 from modulus q to modulus p . Writing that rounded vector as $t \in \mathcal{R}_p^9$ gives the centered public relation

$$As_1 - 32t = e \pmod{q}, \quad e \in [-16, 15]^{9 \times 128}. \quad (1)$$

The public key stores a matrix seed and t_1 ; with $d = 10$,

$$t = 2^d t_1 + t_0 \pmod{p}, \quad (2)$$

where every centered coefficient of the omitted t_0 belongs to $[-511, 512]$. The packed signing key contains s_1 and t_0 , together with the public matrix seed, a signing-randomness seed, and a hash of the public key.

For each signing attempt, the signer samples

$$y \leftarrow [-(\gamma_1 - 1), \gamma_1 - 1]^{6 \times 128}, \quad \gamma_1 = 2^{19},$$

derives a signed sparse challenge c of Hamming weight $\kappa = 31$, and forms

$$z = y + cs_1. \quad (3)$$

The first gate restarts unless

$$\|z\|_\infty < \gamma_1 - \beta_1, \quad \beta_1 = 51. \quad (4)$$

Later gates check a second low-part condition, consistency of decomposed high bits, the magnitude of ct_0 , and the hint weight. The signature transmits z , the hint, and c .

2.1 Separating the written design from submitted-code defects

The submitted ATLAS-128 sampler has two defects relevant to any experiment. Its mask decoder overwrites the first 20-bit value before storing it, so every pair of adjacent mask coefficients is equal. Lu and Liu used that relation to recover s_1 from two signatures and forge a new message; the NGCC report classifies this as an implementation issue [2]. Separately, the submitted $\eta = 16$ secret sampler splits each byte into two four-bit values, so it cannot sample the specified 33-element interval and in fact produces only positive coefficients.

Our faithful experiment replaces those routines with rejection samplers for two independent 20-bit mask values and for a uniform six-bit encoding of $[-16, 16]$. It leaves the key equations, signing gates, packing, challenge generation, and verifier logic unchanged. Thus adjacent-mask cancellation is unavailable, and the target key follows the distribution stated in the specification.

3 Exact first-gate support leakage

Consider one response coefficient and write

$$a = (cs_1)_j, \quad z = y + a, \quad Y = \gamma_1 - 1 = 524287,$$

and

$$C = \gamma_1 - \beta_1 - 1 = 524236.$$

The mask support is $[-Y, Y]$, and the strict norm test in Equation (4) accepts precisely $z \in [-C, C]$.

Lemma 1 (Accepted support). *For fixed a , the accepted response support and its cardinality are*

$$I(a) = [-C, C] \cap [a - Y, a + Y], \quad (5)$$

$$|I(a)| = 1,048,473 - (|a| - 51)_+. \quad (6)$$

Proof. The response must satisfy both $-C \leq z \leq C$ and $-Y \leq z - a \leq Y$, which gives Equation (5). If $|a| \leq Y - C = 51$, the first interval is contained in the second and has $2C + 1 = 1,048,473$ elements. Translating farther removes exactly $|a| - 51$ integers from one endpoint. The maximum possible magnitude $\kappa\eta = 496$ is far below the point at which both endpoints would be truncated, so Equation (6) follows. $\square$

The specification’s constant-width calculation is therefore valid only for $|a| \leq 51$. It separately notes that the parameter-selection constraint forces $\beta_1 \approx \kappa\eta$ to cease being true, but the rejection analysis does not propagate that fact. In particular, the claim that the response distribution is statistically independent of s_1 does not follow for ATLAS-128.

Every observed coefficient gives

$$z - Y \leq a \leq z + Y. \quad (7)$$

Since a is a signed negacyclic linear form in the 128 coefficients of one secret polynomial, Equation (7) is public and linear. The a priori bound $|a| \leq \kappa\eta = 496$ makes most instances vacuous. A nonvacuous lower inequality has the normalized form

$$\langle x, s \rangle \geq b, \quad b \in \{-495, \dots, -51\}. \quad (8)$$

A nonvacuous upper inequality is negated into the same form. At most one such row arises from a response coefficient. In the three-million ideal calibration, approximately 0.647 rows arose per signature.

Remark 2 (Why feasibility is insufficient). *Every bound in Equation (8) is negative, so the all-zero secret satisfies every support inequality. A feasibility solve or an ordinary minimum-norm decoder is consequently directed away from the true secret.*

4 Conditional likelihood and a radial surrogate

Let G be the first-gate event. In an ideal random-oracle model, a fresh challenge is uniform over the signed weight-31 challenge set before the gate. For a candidate s , define

$$P_s(G) = \mathbb{E}_c \left[\prod_{r=1}^6 \prod_{j=0}^{127} \frac{|I((cs_r)_j)|}{2Y + 1} \right]. \quad (9)$$

Proposition 3 (Ideal first-gate candidate likelihood). *Fix an accepted public transcript of challenges and responses. In the ideal first-gate model, the likelihood of a candidate s is zero if any support inequality is violated. Among feasible candidates it is proportional to $P_s(G)^{-N}$, where N is the number of signatures.*

Proof. Before conditioning on G , a particular challenge has candidate-independent probability and every mask vector has probability $(2Y + 1)^{-768}$. Thus a particular feasible pair (c, z) has the same unconditional mass for every candidate; an infeasible pair has mass zero. Conditioning each independent record on G divides its feasible mass by $P_s(G)$. Multiplication over N records proves the claim. $\square$

Proposition 3 says that maximum likelihood selects a feasible candidate with the *smallest* first-gate acceptance probability. Using Lemma 1, its leading relative loss is

$$\frac{1}{1,048,473} \mathbb{E}_c \sum_{r,j} (|(cs_r)_j| - 51)_+. \quad (10)$$

For a random sparse signed challenge, a convolution coefficient is approximately centered Gaussian with variance proportional to the squared coefficient norm of the corresponding secret polynomial. The expectation of the hinge in Equation (10) increases with that variance. This motivates maximizing squared norm over the feasible support polytope.

The norm replacement is a heuristic. Proposition 3 is exact only for the isolated ideal first gate, while genuine signatures have later rejection gates and public hints. We therefore calibrate the surrogate on an exact first-gate simulator and separately test it on genuine signatures. The experiments, rather than a theorem, establish recovery for the two keys reported here.

4.1 A public starting direction

The retained bounds provide a useful first direction without any secret labels. For a normalized row, let $a = \langle x, s \rangle$. Conditional on that endpoint producing a row, the bound b is uniform on the integers

$$-495 \leq b \leq \min(-51, a).$$

Consequently,

$$\mathbb{E}[2b + 495 \mid x, \text{row}] = \min(-51, a). \quad (11)$$

We regress the entirely public target $2b + 495$ on x with a fixed ridge of 10^{-3} and retain only the normalized direction. Although the censoring in Equation (11) prevents this regression from being a direct coefficient estimator, it gives a consistently useful radial start.

5 Public recovery algorithm

The final execution reads only the public transcript and public key. Its continuous stage is fixed for all six secret polynomials.

1. Decode z and c . Convert every nonvacuous instance of Equation (7) to $\langle x, s \rangle \geq b$, discarding the rest.
2. Solve the public ridge regression from Equation (11) and normalize its result to a direction d_0 .
3. Initialize a working set with the 2,000 rows having the largest public bounds. Solve

$$\max_x d_i^T x \quad \text{subject to} \quad \langle x_r, x \rangle \geq b_r, \quad -16 \leq x_j \leq 16.$$

Check the result against every public row and add up to the 2,000 largest violations. Repeat separation until all continuous violations are at most 10^{-6} .

4. Set $d_{i+1} = x / \|x\|_2$ and repeat the radial linear program at most ten times. The working set is retained between iterations.
5. Restrict every integer coefficient to $\{\lfloor x_j \rfloor, \lceil x_j \rceil\} \cap [-16, 16]$. For each polynomial, select the 5,000 rows with least public slack and run a binary integer program maximizing the frozen radial objective. Give each polynomial 20 seconds, validate its incumbent against every exact support row, and add violations as cutting planes. The six response-feasible incumbents form a public baseline $\bar{s}$.

6. Recover t_0 as described in Section 6 and form $t = 2^{10}t_1 + t_0$. Run the public singleton-support lattice ladder below for blocks $r = 0, 1, \dots, 5$ in that order. Accept a candidate only after every collected support inequality and all 1,152 exact LWR residual bounds have been checked with integer arithmetic. If the response-only baseline already passes both checks, it is itself a complete public recovery; the ladder may still be run as prescribed.

The 10^{-6} tolerance is used only for floating-point separation. Integer support validation and the LWR relation use exact integer arithmetic. In an early genuine checkpoint, a 10^{-7} recheck repeatedly classified an active LP boundary as violated at solver precision while adding no new row; the larger tolerance and an explicit no-new-row check remove that numerical stall without using secret information.

5.1 Deterministic LWR block ladder

Fix a block r . The other five polynomials remain equal to $\bar{s}$. In block r , put every coefficient at the lower endpoint of its public floor/ceiling box and let $d \in \{0, 1\}^v$ select the upper endpoint at the v non-singleton coordinates. Select $m = 64$ LWR equations by a fixed public order: enumerate seven-bit coefficient indices $i = 0, 1, \dots, 127$, replace i by its bit reversal, cycle through the nine output polynomials, and take the first 64 rows. After subtracting the fixed contribution, these equations have the form

$$Bd - b = e \pmod{q}, \quad e \in [-16, 15]^m. \quad (12)$$

Set $u = 2d - \mathbf{1} \in \{-1, 1\}^v$ and $h = 2b - B\mathbf{1}$. Equation (12) implies

$$Bu - h = 2e \pmod{2q}. \quad (13)$$

We use the row lattice generated by

$$\mathcal{B} = \begin{pmatrix} 2qI_m & 0 & 0 \\ B^T & \lambda I_v & 0 \\ h^T & 0 & M \end{pmatrix}, \quad \lambda = 18, \quad M = 1. \quad (14)$$

If this block contains the needed corrections, the lattice contains

$$(2e, \lambda u, -M), \quad (15)$$

obtained by combining the secret rows with coefficients u , subtracting the last row, and adding suitable $2qI_m$ rows. For the successful block, the embedding dimension is 187 and the vector in Equation (15) has squared norm 70,329.

Lattice reduction does not force the middle coordinates to be binary. We run LLL with $\delta = 0.99$, followed when necessary by at most two BKZ-10 tours and at most two BKZ-20 tours using a 256-bit MPFR Gram–Schmidt representation. We inspect a reduced row only if its final coordinate is exactly $\pm M$ and every middle coordinate is exactly $\pm \lambda$. After normalizing its sign, those middle coordinates define u and hence $d = (u + \mathbf{1})/2$. All other reduced rows are ignored. A decoded candidate still has no standing until it passes the complete support and LWR validations.

Both reported runs fix the ladder to the six singleton supports. On the calibration key, blocks 0, 1, and 2 completed LLL, BKZ-10, and BKZ-20 without a binary embedding row; block 3 succeeded after BKZ-10. On the prospective key, the integer baseline already passed every public check. The prescribed ladder was nevertheless run unchanged and rediscovered that identical candidate for block 0 after BKZ-10. A larger deterministic ladder could continue to block pairs if all singletons failed; that extension was neither needed nor tested here.

The optional post-experiment comparison with the generated secret is absent from every regression, objective, row choice, coefficient box, lattice, stopping rule, and public validation above.

6 Recovering the omitted low part t_0

The public key contains t_1 but Equation (1) needs the full rounded product t . The ATLAS hint supplies bounded public observations of t_0 . From a signature, the verifier can compute

$$a = \text{Round}(Az) - c(2^d t_1) \pmod{p} \quad (16)$$

and use the public hint to reconstruct the intended high part w_1 . Centering the corresponding low representative gives an observable u satisfying

$$u = ct_0 + r_0, \quad |r_0| \leq 8183. \quad (17)$$

For each of the nine polynomials, we solve the public minimax problem

$$\min_{x \in [-511, 512]^{128}} \max_i |(c_i x - u_i)| \quad (18)$$

by a cutting-plane LP and round the solution. A nearby-integer completion is accepted only after every coefficient of every observation satisfies Equation (17).

On each target used below, 20,000 signatures recovered all 1,152 coefficients. On the calibration and prospective keys, respectively, the continuous RMSE values were 0.116057 and 0.120930, and the maximum errors were 0.438380 and 0.464272. Both rounded vectors had maximum public residual 8,183 and were returned unchanged by the independent integer pass. Prospective collection took 123.9 seconds and the minimax solve took 260.9 seconds wall time with 482 MiB peak RSS.

This high-level partial-key recovery is analogous to the recovery of Dilithium’s omitted t_0 by Azevedo Oliveira, Calle Viera, Cogliati, and Goubin [3]. We do not claim the t_0 step alone as a new attack. Its role here is to expose the full public LWR validation relation once the separate first-gate channel recovers s_1 .

7 Calibration and genuine-transcript experiments

7.1 Ideal first-gate calibration

Our ideal simulator samples a uniform signed weight-31 challenge, weights it by its exact first-gate acceptance probability under the calibration secret, and then samples every response coefficient uniformly from $I((cs_1)_j)$. It deliberately omits every later gate. The public recovery algorithm does not receive the calibration secret.

Table 1: Ideal first-gate calibration on the target secret. “Neighbors” counts valid one-coordinate ± 1 alternatives that satisfy every support row.

Signatures	Retained rows	Neighbors	Radial RMSE	Rounded exact
100,000	64,327	807/1,497	12.4346	25/768
1,000,000	646,782	18/1,497	0.877129	344/768
3,000,000	1,939,568	0/1,497	0.238428	734/768

At three million ideal records, the maximum continuous coefficient error was 0.857099, placing every true integer inside its public floor/ceiling box. The six 20-second integer programs recovered all 768 coefficients. The assembled result passed all 1,152 public LWR checks with residual range $[-16, 15]$. Simulation took 511.8 seconds, radial recovery took 182.2 seconds with 1.36 GiB peak RSS, and integer completion took 127.9 seconds with 1.48 GiB peak RSS.

7.2 Experimental protocol

Both genuine experiments used a four-core Intel Core i5-6500 desktop with 16 GiB RAM. Distinct deterministic 64-byte key-generation seeds fix the two ATLAS-128 keys for reproducibility. For each key, query messages are distinct 16-byte encodings of the counters $0, \dots, 2,999,999$ under fixed public byte pads. Four processes handle disjoint consecutive counter ranges. Every process independently derives the same target key, calls the faithful signer, parses the ordinary signature, and streams only the nonvacuous public rows.

For each key, the 20,000-record hint experiment uses the first 20,000 query messages. Because ATLAS signing is deterministic, these are repeated records of queries already counted in the three-million set, rather than additional distinct messages. The public keys embedded by the hint and response collectors match byte for byte.

The first experiment served as calibration during method development. For the second experiment, the complete procedure, parameters, block order, and stopping rules were frozen before the next unused deterministic key identifier was selected. The freeze manifest has SHA-256 digest

9bebb94f64c14906f1284b25293a32d65bb86504c8bc96f95300330b0545a18c

The signing collectors necessarily held the target secret internally to generate oracle responses, but their optional calibration outputs were disabled and the recovery pipeline received only public artifacts. Public recovery, complete public validation, equivalent-key construction, and fresh-message forgery all finished before the target secret was exported, inspected, or supplied to any recovery or evaluation step. A manifest of those immutable public artifacts was then sealed with digest

f89c9e1433ea6ecca1ae963b40bee4992e5eaa09c56587c2419983751973ecfd

Only afterward did a separate calibration step regenerate the deterministic secret and score the outputs.

7.3 Calibration checkpoints and final recovery

At 100,000 signatures, both laws are underdetermined. At one million, the genuine signer tracks the ideal calibration closely:

Table 2: Public radial estimator on ideal and genuine transcripts.

Transcript	Signatures	RMSE	Correlation	Rounded exact
Ideal first gate	100,000	12.4346	0.3312	25/768
Genuine signer	100,000	11.6469	0.4089	34/768
Ideal first gate	1,000,000	0.877129	—	344/768
Genuine signer	1,000,000	0.803564	0.996706	361/768

The genuine one-million transcript contains 644,828 rows. Its public solve took 93.4 seconds while four further collection processes ran and peaked at 518 MiB RSS. Afterward, a calibration-only audit found 29 of 1,497 nearest alternatives still feasible, compared with 18 in the ideal model.

The final genuine transcript contains 1,934,304 rows. Its public radial point has RMSE 0.244287 only in the later calibration comparison; the recovery run did not know that score. Direct rounding gives 731 correct coefficients. That comparison also shows that the true integer lies in all 768 public floor/ceiling boxes. Six response-only integer programs produce a baseline satisfying every support row. This baseline fails the public LWR check, with none of 1,152

residuals in $[-16, 15]$. Only afterward, a calibration comparison finds 745 correct coefficients: blocks 0, 1, 2, 4, and 5 are exact, while block 3 has 23 one-step errors.

The immutable lattice-ladder output passes every support row and all 1,152 LWR checks. A separate calibration comparison then confirms 768 of 768 coefficients, with maximum error zero. The public singleton supports 0, 1, and 2 consume their complete reduction schedules; support 3 succeeds after BKZ-10. The ladder takes 480.3 seconds and peaks at 590 MiB RSS. Its output, serialized as six lines of 128 signed decimal integers with single spaces and newline terminators, has SHA-256 digest

```
aacacca14b8fb842670ea85e939adc0229f04a212aec34d57af50ba7150c4669
```

This execution itself did not load the target secret, but the algorithm had been developed while that secret was available. In particular, the single-block completion was designed after a diagnostic localized the baseline’s 23 errors to block 3.

7.4 Prospectively frozen second-key recovery

The second transcript contains 1,931,974 support rows and has SHA-256 digest

```
355829fab9c9b869fa0d28eebd0e93258d276f5be69149228aeeecbaf9e7bec4
```

The public radial point was written before calibration. The later comparison measured RMSE 0.249334, maximum error 0.833201, correlation 0.999676153, and 728/768 correct coefficients after direct rounding. Every true coefficient lay in its public floor/ceiling box.

All six response-only integer incumbents passed every support inequality. Their assembled baseline also passed all 1,152 public LWR checks with residual range $[-16, 15]$. The prescribed lattice ladder was nevertheless run without change; block 0 succeeded after BKZ-10 and returned the identical vector. Before calibration, this vector had already formed an equivalent key and produced an accepted fresh-message signature. The later comparison found exact s_1 recovery, 768/768, and exact t_0 recovery, 1,152/1,152. A post-experiment audit found that none of 1,481 valid one-coordinate ± 1 alternatives satisfied all support rows; every alternative violated at least two. Counting the separate 20,000-record hint recollection and every public recovery stage, the measured prospective run took about 5,817 seconds (97.0 minutes); a unified transcript collector could avoid the 123.9-second hint recollection. The serialized recovered s_1 has digest

```
c5111d7b13af090288d583081f1f493747bd28283042069e9d40ccffe8c73375
```

Table 3: Genuine-transcript recovery on the calibration and prospectively frozen keys. Calibration metrics in the lower rows were computed only after the corresponding public artifacts were immutable.

Quantity	Calibration key	Prospective key
Chosen-message signatures	3,000,000	3,000,000
Retained support rows	1,934,304	1,931,974
Collection wall time (s)	5,207.6	5,064.7
Aggregate signatures/s	576.1	592.3
Continuous radial RMSE	0.244287	0.249334
Continuous maximum error	0.884067	0.833201
Correct after direct rounding	731/768	728/768
Nearest alternatives surviving	0/1,497	0/1,481
Radial solve wall time (s)	86.2	160.0
Integer baseline wall time (s)	125.1	126.1
Baseline exact coefficients	745/768	768/768
Baseline public LWR check	failed	passed
Ladder stopping point	block 3, BKZ-10	block 0, BKZ-10
Ladder wall time (s)	480.3	75.6
Peak recovery RSS	1.45 GiB	1.42 GiB
Final exact s_1 coefficients	768/768	768/768
Public LWR residual range	$[-16, 15]$	$[-16, 15]$

8 Equivalent signing keys and fresh-message forgeries

Recovering s_1 and t_0 does not reveal the original signing-randomness seed K . That seed is not needed for an equivalent key. From the public key we parse the public matrix seed ρ and compute $tr = H(pk)$. We choose a fresh K' , then pack

$$sk' = (\rho, K', tr, s_1, t_0) \quad (19)$$

in the specified 2,128-byte secret-key format.

The forgery message is an 80-byte fixed string selected before the prospective run. Every signing-oracle query has length 16, so freshness follows immediately, independently of message contents. For each key, the faithful signer using sk' returns a 2,081-byte detached signature, and the faithful verifier using the corresponding original public key accepts it. For reproducibility, the SHA-256 digests are

Calibration signature:	cda3afd63f3f1149b2b1ffe7746ae4db045dbf77bdf88b1e92eda5ebd838c29
Calibration equivalent key:	fd3bdc055d0bb376031bf556721c371c7e0c2b6f56850fc6aaf6b3509e2e499b
Prospective signature:	b4a98b0913b409c8ded21762cf0bd69b4d08aef19fba4e7f2c4c85a6869da2bb
Prospective equivalent key:	32130947f97b0982c4e89124b7f0f076f926042810f9e35bc579d610469f49d9

This establishes exact s_1 recovery followed by equivalent signing-key construction and a successful fresh-message forgery for each tested key. It does not recover either original seed K .

9 Scope, limitations, and interpretation

The interval calculation in Lemma 1 and every support row in Equation (8) are exact consequences of the written ATLAS-128 parameters. The constant-width rejection argument in the specification is therefore false whenever $|(cs_1)_j| > 51$. This is a design issue, independently of the submitted repeated-mask implementation flaw.

The efficient decoder is partly heuristic. Proposition 3 is exact for the isolated ideal first gate, but the squared-norm surrogate, Gaussian convolution model, and behavior under later signing

gates are not proved. The two genuine experiments show that they work on both reported targets. They do not establish a key-averaged success probability or a three-million worst-case bound.

The procedure was developed retrospectively on the calibration key. A secret-based diagnostic localized its 23 baseline errors to block 3 and motivated the singleton ladder. The final calibration execution used no secret input and accepted only public checks, but its development could have overfit that target. For the second experiment, we froze and hashed the complete procedure before generating a distinct deterministic key. No ground-truth secret file existed during recovery or forgery, the public result was hashed before calibration, and the frozen procedure succeeded unchanged.

A formal EUF-CMA advantage is a probability over fresh key generation and the adversary’s randomness, which two successful experiments cannot estimate. The concrete result is exact recovery and a fresh-message forgery on both tested written-design keys, with one execution prospectively frozen.

The experiments cover ATLAS-128 only. Other parameter sets have different n , ℓ , κ , η , and β_1 , so neither the sample count nor the same decoder should be extrapolated without a separate analysis.

10 Conclusion

ATLAS-128’s first rejection gate intersects a translated response interval with a finite mask support. Because β_1 is much smaller than the allowed challenge-product magnitude, that intersection is secret dependent. The resulting support half-spaces do not identify the key by feasibility, but their conditional likelihood favors large-radius feasible secrets. On both reported keys, public radial LPs narrow every coefficient to adjacent integers, public integer completion recovers a response-feasible baseline, and the complete LWR relation identifies exact s_1 from three million genuine signatures. The first key needs the public Kannan-embedding ladder; the prospective second key is already exact after integer completion, and the prescribed ladder independently rediscovers it. Public hints recover t_0 , and equivalent keys sign fresh messages accepted by the original verifiers.

The result warrants independent human review and external reproduction. It also shows that rejection-sampling interval counts must include the original sampler boundary: translation invariance of an unconstrained interval is not enough.

References

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